AVAILABLE-STAGE VALIDATION | LOCATION TABLES PENDING

SpectralBridge Drone Pipeline: Three-Site End-to-End Validation

Decision: available source stages completed; full end-to-end validation remains incomplete. Gordon, Goldhill and Ruby original-package imagery was discovered in CyVerse, downloaded with remote checksum verification and prepared at 10 cm. Existing production code created working HDF5 and ENVI products and standard QA. New camera-location tables were explicitly deferred by the user.

All three requested production runs, with both corrections and translation enabled, stopped at the physical-calibration gate. The original integer image values cannot yet be certified as physical reflectance. This is a documented scientific limitation, not evidence that a correction algorithm failed.

Verified available work	Result
Source packages / transferred inputs	3 packages / 48 files
Prepared imagery	10 spectral bands at 10 cm for every site
Spatial polygon support	302 half-diameter polygons; 200,363 indexed pixels
Physical corrections / translation / approved libraries	Not completed; no scientific acceptance bypass
Production uploads	Not attempted: no fully validated completed site

This report evaluates source handling and numerical preparation. It cannot answer whether BRDF or topographic correction introduces harmful artifacts because no eligible corrected products were generated in this run.

Software: 2.3.0rc2; preparation branch codex/prepare-2.3.0rc2. Baseline commit bb3da46bea0a4295ca3152a9f607ed17140575f1 with recorded working-tree edits. Earlier Goldhill research products are not relabeled as results of this integration test.

What was discovered

The intake listed actual contents under /iplant/home/shared/earthlab/macrosystems/field-data/uas-packages. It selected original-package ORTHO exports and retained associated reports, metadata and terrain alternatives. It did not use historical normalized 10 cm imagery as the radiometric source.

Site	Selected orthomosaic	Native size
Gordon	AOP-GORDON-8-14-23-M300_ORTHO.tif	2765 x 3918 pixels
Goldhill	AOP-GOLDHILL-08-14-23_ORTHO.tif	2646 x 3747 pixels
Ruby	AOP-RUBY-08-14-23-ORTHO_CLIPPED.tif	1242 x 2354 pixels

Ruby coverage qualification: the only selected original-package orthomosaic is ORTHO_CLIPPED. It is a spatially clipped export; this test does not establish that an unclipped original was available.

Gordon uses a different date/name spelling from Goldhill. Discovery handles that variation and case-insensitive site names. More than one acceptable ORTHO candidate produces an ambiguity report instead of an automatic choice.

The associated SD1/SD2 image-metadata tables were preserved as source metadata. They were not substituted for the new location tables. Point-cloud archives were not downloaded because the available stages do not require them.

Remote MD5 checksums were compared using their actual algorithm. Logical Data Store paths, byte sizes and local SHA-256/MD5 records are preserved in each source manifest. Physical server replica paths are never used as inputs.

Source validation and preparation

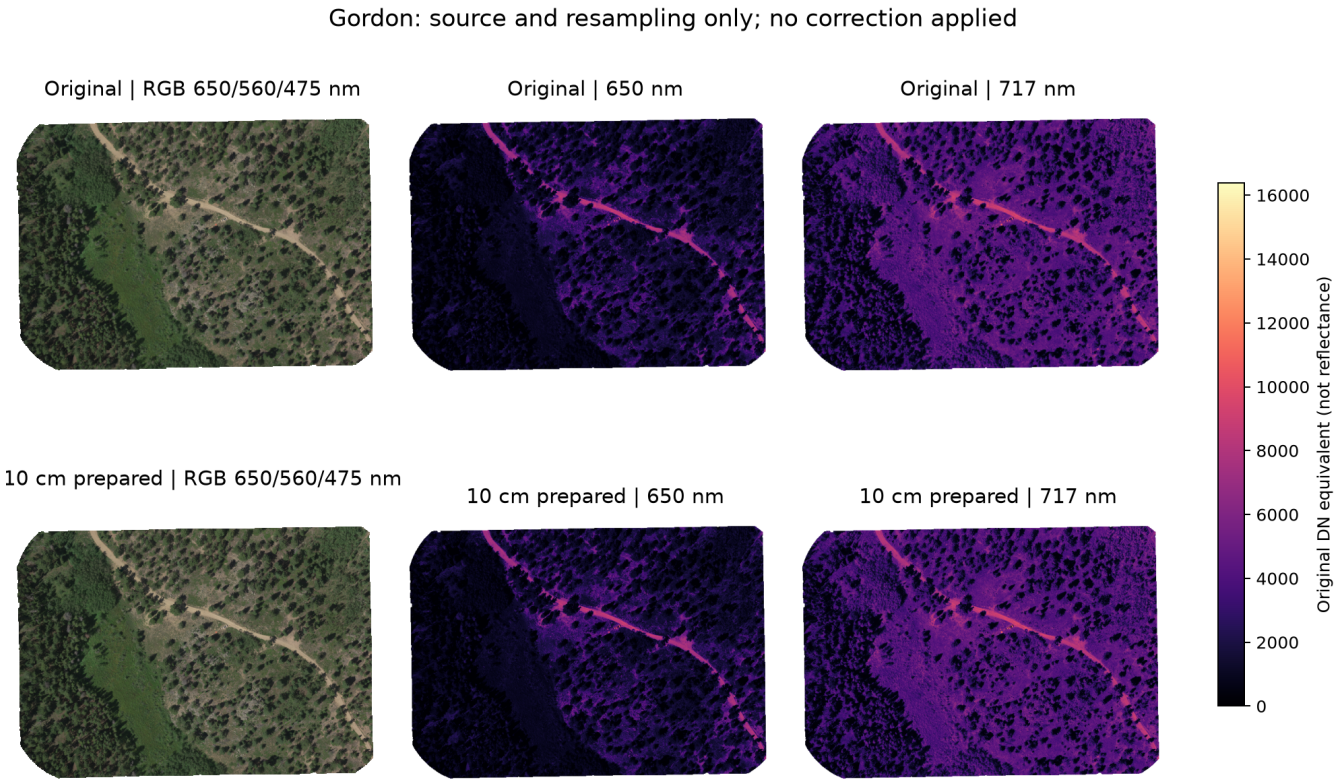
Property	Finding
Raster geometry	All three: projected NAD83 / UTM zone 13N, north-up, east-right, unrotated
Spectral contract	11 UInt16 bands: 10 signal bands plus declared alpha band 11
Wavelengths	444, 475, 531, 560, 650, 668, 705, 717, 740, 862 nm
Wavelength evidence	Established package convention; TIFF files have no independent wavelength declarations
Native pixel size	Gordon 8.063 cm; Goldhill 8.443 cm; Ruby 8.315 cm
Prepared grid	10 cm; existing recorded grids used only for spatial comparison support
Numerical conversion	stored = DN x 10000 / 65535; offset 0; no min/max adjustment
Physical conversion	Unknown: numeric range alone does not certify reflectance

Bilinear resampling excludes source alpha/NoData; the footprint mask is resampled with nearest neighbor. Individual missing spectral bands remain missing without erasing observed bands. Zero is retained unless explicitly declared missing. Source files remain unchanged.

Terrain warning: Goldhill has distinct DTM.tif and dtm.tif objects. Stable separate local directories now prevent case collisions. The uppercase terrain raster opens; the tiny lowercase file does not open as a valid raster despite matching its remote checksum. It was retained and marked invalid, not repaired or adopted.

Preparation works one spectral band at a time, with source/target masks in memory. RAM scales with raster area; it is not a window-bounded resampler. HDF5 and ENVI conversion continue to use the existing chunked pipeline.

Gordon: original and prepared imagery

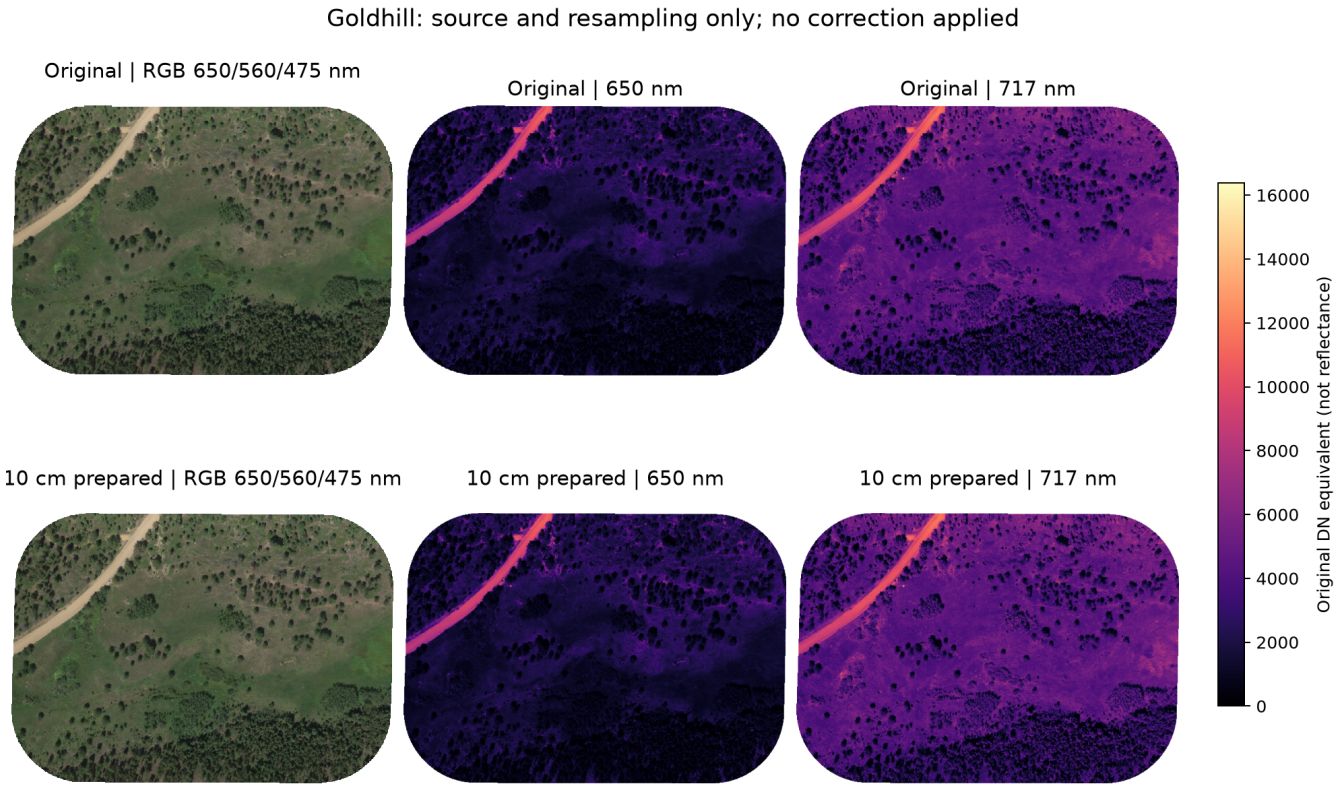


Top: native original. Bottom: 10 cm preparation. RGB uses 650/560/475 nm and a fixed square-root stretch. Red (650 nm) and red-edge (717 nm) maps share a fixed 0-16,384 DN-equivalent scale. Both rows share map limits. Display sampling does not alter analytical statistics.

Band	Valid pixels	Median DN equivalent	1st / 99th percentile
650 nm	5,898,401	1,127.5	92.4 / 7,198.9
717 nm	5,898,401	3,487.0	305.7 / 7,888.4

The broad scene structures remain visible after preparation. This side-by-side check is descriptive: it does not establish physical calibration, eliminate all possible source seams, or validate correction factors. No topographic-only, BRDF-only or combined corrected result exists for this integration run.

Goldhill: original and prepared imagery

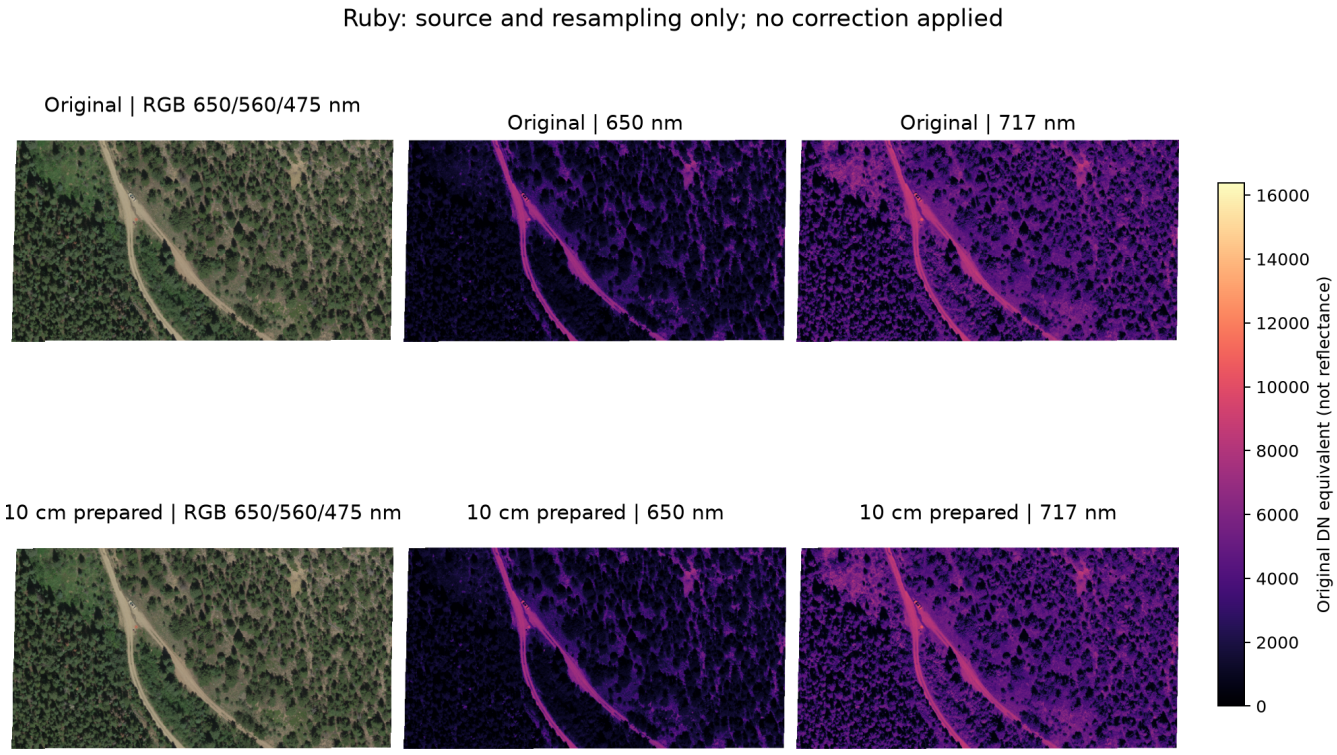


Top: native original. Bottom: 10 cm preparation. RGB uses 650/560/475 nm and a fixed square-root stretch. Red (650 nm) and red-edge (717 nm) maps share a fixed 0-16,384 DN-equivalent scale. Both rows share map limits. Display sampling does not alter analytical statistics.

Band	Valid pixels	Median DN equivalent	1st / 99th percentile
650 nm	6,683,944	1,488.9	109.9 / 8,929.4
717 nm	6,683,944	4,302.3	386.3 / 9,724.1

The broad scene structures remain visible after preparation. This side-by-side check is descriptive: it does not establish physical calibration, eliminate all possible source seams, or validate correction factors. No topographic-only, BRDF-only or combined corrected result exists for this integration run.

Ruby: original and prepared imagery

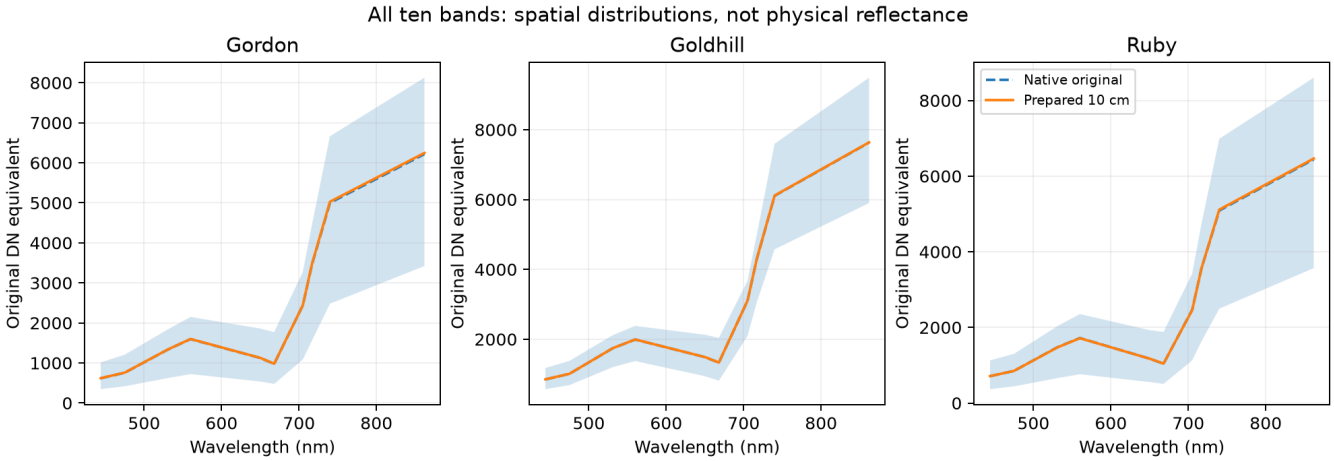


Top: native original. Bottom: 10 cm preparation. RGB uses 650/560/475 nm and a fixed square-root stretch. Red (650 nm) and red-edge (717 nm) maps share a fixed 0-16,384 DN-equivalent scale. Both rows share map limits. Display sampling does not alter analytical statistics.

Band	Valid pixels	Median DN equivalent	1st / 99th percentile
650 nm	1,993,472	1,174.5	92.1 / 7,308.8
717 nm	1,993,472	3,558.4	342.9 / 8,193.9

The broad scene structures remain visible after preparation. This side-by-side check is descriptive: it does not establish physical calibration, eliminate all possible source seams, or validate correction factors. No topographic-only, BRDF-only or combined corrected result exists for this integration run.

All-band and cross-site diagnostics



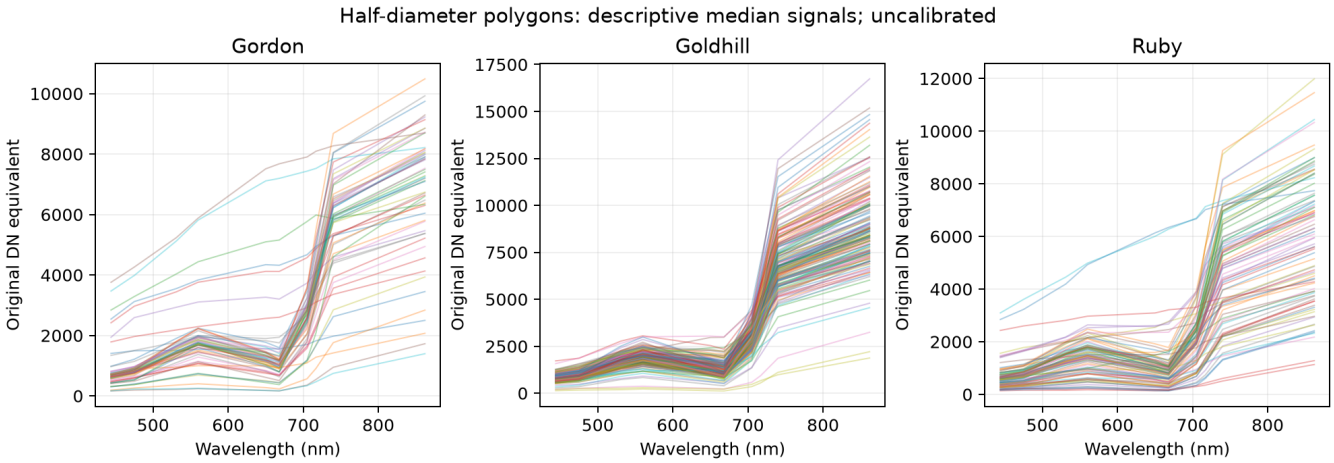
Solid lines show the median of every finite prepared pixel in each band; dashed lines show native-source medians. The shaded range is the prepared interquartile range. Prepared values are converted back to original DN equivalent solely to compare numeric distributions. Site differences must not be interpreted as calibrated sensor or vegetation differences.

Site	10 cm raster support	Valid 650 nm pixels	Prepared/native median change
Gordon	2088 x 2948	5,898,401	+0.22%
Goldhill	2238 x 3165	6,683,944	+0.06%
Ruby	1035 x 1960	1,993,472	+0.05%

The accompanying CSV contains all ten bands, both grids, valid/missing counts, minima, maxima, percentiles and 99th-percentile adjacent-pixel jumps in both directions. Native and prepared populations differ through resampling and spatial support; changes in their summaries are not a pixel-by-pixel radiometric calibration residual.

Artifact conclusion: harmful correction artifacts are not evaluable. A zero raw-versus-identity difference in standard QA simply confirms that no correction was requested in the preparation run.

Polygon support without calibration claims



Site	Half-diameter polygons	Indexed pixel centers
Gordon	58	32,930
Goldhill	158	144,749
Ruby	86	22,684

The existing polygon-index utility retains polygon attributes, source references and pixel-center coordinates. Each curve is a polygon median signal. These descriptive DN summaries are explicitly marked DESCRIPTIVE_DN_NOT_APPROVED_LIBRARY. They are not approved reflectance libraries, calibrated vegetation signatures or translated products.

The source is the established half-diameter GeoPackage. Full-diameter polygons were not substituted. Polygon IDs, recorded site/cover/PFT attributes and provenance remain attached to the index and compact diagnostic table.

The actual production result

Stage	All three sites
Discovery / download / source validation	Completed for selected orthomosaics; terrain alternatives assessed
10 cm preparation / working H5 / ENVI	Completed; restart reuse exercised
Standard preparation QA	Generated; physical scientific status NOT_EVALUATED
Topographic correction	Requested; blocked before application by unverified radiometry
BRDF correction / reconstructed geometry	Requested; same earlier gate; new location tables deferred
Combined correction / translation	No eligible corrected input; not produced
Approved spectral libraries	Not produced; existing acceptance requirements preserved
Production upload / remote output verification	Not attempted because no site completed the required production validation

Exact first blocking reason: Radiometry UNSUPPORTED: authoritative physical scale, offset, units and provenance required (Original UInt16 DN; fixed numerical encoding; physical calibration unavailable).

No viewing-angle maps, kernel conditioning, illumination support or correction-factor distributions are reported as observed results. These depend on inputs/stages that did not complete. The standard blocked-run QA summary records the reason; source-only QA figures remain separately scoped.

The new location tables are deferred by user instruction. Their future validator requires finite camera positions, unique identifiers, timestamp syntax, explicit projected metre CRS and overlap with the raster. It does not auto-relabel CRS or certify vertical datum, clock zone, orientation or mosaic contributors.

Reproducibility and next steps

Integration defects repaired: actual JSON listing/checksum handling; stable case-distinct local staging; source-only QA scientific status; restart of an explicitly uncorrected identity product without falsely requiring physical correction provenance. Requested correction gates remain active.

Checksummed source/configuration/output preparation records support restart. The real pipeline reuses working HDF5 and ENVI stages. All 48 staged inputs were checked against their remote identities and exercised through offline reuse with downloads forbidden. Exact prepared-to-ENVI comparisons check every finite value and missing-data position in bounded windows.

Site	Verified inputs	Exact finite ENVI values
Gordon	14	58,984,010
Goldhill	15	66,839,440
Ruby	19	19,934,720

BLOCKER for the full objective: no scientifically eligible corrected production chain and therefore no completed production upload. **IMPORTANT:** unreadable lowercase Goldhill terrain alternative and post-freeze source changes requiring replacement release-artifact validation. **DEFER:** user-pending camera tables, independent physical calibration and actual contributor validation; these are scientific/input limitations, not automatically algorithm defects.

To resume: supply validated location tables and an accepted radiometric scale/offset with provenance; align and validate terrain/geometry through existing contracts; rerun both corrections, translation and approved-library extraction; evaluate all correction stages and upload only a completed, validated site to a new Data Store directory. Verify remote checksums before declaring end-to-end success.

The archived RC2 wheel/sdist predate these integration edits. Their earlier test results apply to those exact bytes. A replacement candidate must be built and tested before publication. No release, tag, merge or production upload was performed here.

Evidence: provenance/drone-three-site-integration/2026-11-04. Large local inputs, prepared TIFFs and pipeline outputs: archive/local/drone-three-site-integration/2026-11-04. See DRONE_THREE_SITE_INTEGRATION_REPORT.md for commands, validation results and an inventory of deliverables.